

Rhutika Patil

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PROFESSIONAL SUMMARY

Bioinformatics MS graduate (NC State, 2026) with one year of healthcare IT industry experience and four research roles. Builds reproducible data pipelines in Python, R, and SQL, and has modeled 1M+ healthcare records under HIPAA, HL7, and FHIR standards. Experienced in researching and comparing technologies, building proof of concept workflows, and presenting findings to technical and non technical audiences.

EDUCATION

North Carolina State University | Master of Science, Bioinformatics

Aug 2024 to Aug 2026

Savitribai Phule Pune University | B.E., Electronics and Telecommunication

Aug 2018 to Jul 2022

SKILLS

Programming and Data: Python (pandas, NumPy, scikit-learn, Matplotlib), R (ggplot2, RMarkdown, Bioconductor), SQL (PostgreSQL, MySQL), Bash, ETL and data quality

Analytics and ML: statistical modeling, hypothesis testing, anomaly detection, classification, dimensionality reduction (PCA), clustering, multivariate analysis

Genomics and NGS: RNA-seq, scRNA-seq, Tag-seq, Seurat, DESeq2, edgeR, FastQC, MultiQC, HISAT2, STAR, Salmon, GATK, Bowtie, BLAST

Infrastructure and BI: Linux, HPC/SLURM, Docker, Git/GitHub, Galaxy, Power BI, Tableau, Excel | **Standards:** HIPAA, HL7, FHIR

PROJECTS

Medicare Provider Fraud Detection (Proof of Concept) | Python, SQL, scikit-learn

Sep 2026

- Built provider risk profiles from 558K Medicare style claims using SQL, then compared statistical rules, Isolation Forest, and Logistic Regression for payment anomaly detection.
- Logistic Regression caught 71% of fraud labeled providers while reviewing only 10% of providers (PR AUC 0.74 vs 0.09 random); produced a ranked review list with plain English reasons for each flag.

EXPERIENCE

Bioinformatics Consultant | North Carolina State University

Jan 2026 to Jun 2026

- Engineered end to end RNA-seq and Tag-seq pipelines in R and Python, processing 50+ libraries across multiple species and reducing manual preprocessing time by 40%.
- Built reproducible workflows that sped up cross species statistical analysis by 30% and quantified expression across 20,000+ genes to guide follow up experiments.
- Wrote user documentation and workflow guides so team members with varying technical backgrounds could run the pipelines.

Bioinformatics Research Analyst | North Carolina State University

Aug 2025 to Dec 2025

- Integrated high dimensional plant physiology and environmental data from 100+ field observations in R to model drought response.
- Modeled trait and environment relationships with regression and multivariate statistics, improving the predictive reliability of drought tolerance indicators by 25%.

Graduate Research Assistant | NC State College of Veterinary Medicine

May 2024 to Aug 2025

- Built RNA-seq and scRNA-seq pipelines in R and Seurat for ~90 GB of sequencing data from 15 samples, testing analysis choices on small subsets before full runs.
- Automated QC, alignment, and differential expression workflows, cutting manual processing time by 40%.
- Compared the strengths and tradeoffs of Seurat, DESeq2, and edgeR, and documented recommendations before choosing the final analysis stack.
- Presented findings and proposals to interdisciplinary stakeholders, guiding follow on research decisions.

Graduate Research Assistant | Auburn University

Jan 2024 to Apr 2024

- Analyzed bulk RNA-seq profiles from 6+ samples to characterize host pathogen interactions, interpreting pathway level immune and inflammatory responses.

Bioinformatics Software Engineer | CitiusTech

Jul 2022 to Aug 2023

- Modeled 1M+ healthcare records in Python and R to produce population level clinical insights, working with cross functional research teams.
- Designed data validation and transformation pipelines compliant with HIPAA, HL7, and FHIR standards across interconnected clinical systems and relational databases.
- Researched clinical data platforms, data flows, and integration points; wrote technical documentation and use case analyses comparing candidate tools.
- Built Power BI and Tableau dashboards combining clinical and operational metrics, and automated ETL pipelines for clinical datasets.

CERTIFICATIONS

Google Data Analytics Professional Certificate | Machine Learning Specialization (Coursera)